Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for:
Sepsis in Obstetric Patients

Eastern and Northern Services

Summary

This guideline outlines the management of sepsis in Obstetric Patients.

Key Points

- NICE defines sepsis as a life-threatening organ dysfunction due to a dysregulated host response to infection.
- Sepsis can develop quickly and cause significant morbidity and mortality. Therefore the OB Sepsis risk assessment tool, found in the Sepsis tab on the electronic patient record, should be completed and frequent monitoring is of paramount importance
- Use of the Sepsis Six care bundle will provide a structured approach to the initial assessment and management of sepsis
- Antibiotics should be given within one hour, please consult with the Microguide, found on the HUB, for the current antibiotic recommendations.
- A multi-disciplinary approach is essential to provide good quality care.

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1.0 INTRODUCTION

Sepsis is a life-threatening organ dysfunction due to a dysregulated host response to infection. Suspected sepsis is used to indicate people who might have sepsis and require face-to-face assessment and consideration of urgent intervention.

In the triennial report by CMACE 2006-2008, sepsis was the leading direct cause of maternal death. The most recent report by MBRRACE-UK 2019-2021 showed that sepsis now ranks as the fifth most common cause overall of maternal death, having ranked sixth in the previous report. The cause of death from sepsis can be further divided into direct (genital tract source) and indirect (predominantly respiratory source).

Not to be under-estimated is the compounding impact sepsis can have on other obstetric complications such as haemorrhage and thromboembolic disease. Young and healthy women can maintain normal physiology until the late stages of disease, hence the need for rapid assessment and instigation of treatment if sepsis is suspected.

1.1 NICE Sepsis Guideline and MBRRACE Recommendations

This guideline advises that women who are pregnant, have given birth or had a termination of pregnancy or miscarriage in the past six weeks are in a high-risk group for sepsis. In particular, women who:

- have impaired immune systems because of illness or drugs
- have gestational diabetes or diabetes or other co-morbidities
- needed invasive procedures (for example, caesarean section, forceps delivery, removal of retained products of conception)
- had prolonged rupture of membranes
- have or have been in close contact with people with group A streptococcal infection, for example, scarlet fever
- have continued vaginal bleeding or an offensive vaginal discharge
- Infection must be suspected and actively ruled out when a recently delivered woman has persistent vaginal bleeding and abdominal pain.

1.2 Sources of Sepsis in Obstetrics

- Urinary tract infection, pyelonephritis
- Premature pre-labour rupture of membranes, prolonged rupture of membranes
- Chorioamnionitis
- Septic miscarriage, endometritis following delivery or termination
- Respiratory tract infection – influenza, Covid 19 or pneumonia
- Wound infection
- Mastitis

2.0 ASSESSMENT OF SEPSIS

A patient may trigger a sepsis risk assessment from a pyrexia in labour or from an initial presentation to a routine antenatal check, clinic appointment or attendance at the maternity triage.

Take into account the normal physiological changes in labour when thinking about the possibility of sepsis, for example, increased maternal pulse rate.

Complete the OB Sepsis risk assessment tool

Perform a full set of baseline observations as below and document on the MEWS Vital Signs in Flowsheets on the electronic patient record. The ongoing frequency is as per NICE guidance.

Suspected sepsis – concern insufficient for antibiotics						
Heart Rate	**Blood Pressure**	**Respiratory Rate**	**Oxygen Saturation**	**Temperature**	**Level of Consciousness (AVPU)**	**Urine Output**
Hourly	4-hourly OR hourly in the 2nd stage	4-hourly	4-hourly	Hourly	Hourly	Record output

Sepsis or suspected sepsis – ON antibiotics						
Heart Rate	**Blood Pressure**	**Respiratory Rate**	**Oxygen Saturation**	**Temperature**	**Level of Consciousness (AVPU)**	**Urine Output**
Continuous or at least every 30 minutes	Continuous or at least every 30 minutes	Continuous or at least every 30 minutes	Continuous or at least every 30 minutes	Hourly	Every 30 minutes	Record output, hourly if catheterised

3.0 MANAGEMENT OF SEPSIS: SEPSIS SIX CARE BUNDLE:

OXYGEN	Administer 100% oxygen at 15L/min via non rebreathe mask Aim to keep saturations > 94%

↓

CULTURES	• blood* • urine • vaginal (high and/or low) • stool • wound/placenta • breastmilk • sputum • throat swab (bacterial and viral) • nasal swab (influenza) • any other relevant *Aim to obtain blood cultures prior to administering antibiotics but do not delay

↓

BLOODS	Insert 16G cannula and take bloods • venous lactate* • full blood count • urea and electrolytes • C-reactive protein • Coagulation screen • Glucose • Liver function tests • Serial serum lactates to monitor response to treatment *Venous lactate: If venous lactate raised, recheck with arterial sample. Discuss with critical care if arterial lactate ≥ 4mmol/L. Continue to check

↓

ANTIBIOTICS	Give intravenous antibiotics within one hour Consider allergies Use the Microguide, for the current antibiotic recommendations. Target antibiotics to source of sepsis if this is clear, otherwise use broad spectrum antibiotics Order urgent inpatient microbiology consult for complex cases

↓

FLUIDS	Give intravenous fluids If lactate ≥ 2mmol/L give 500mL stat If hypotensive or lactate ≥ 4mmol/L can repeat boluses up to 30 mL/kg (e.g. 2 L for a 70 kg woman) Target mean arterial BP >65mmHg Extreme caution if woman has pre-eclampsia: discuss with anaesthetist

↓

URINE OUTPUT	Urinary catheter & hourly measurement Strict fluid balance recorded in flowsheets on the electronic patient record Target >0.5ml / kg / hour

4.0 MANAGEMENT OF SEPSIS: OTHER

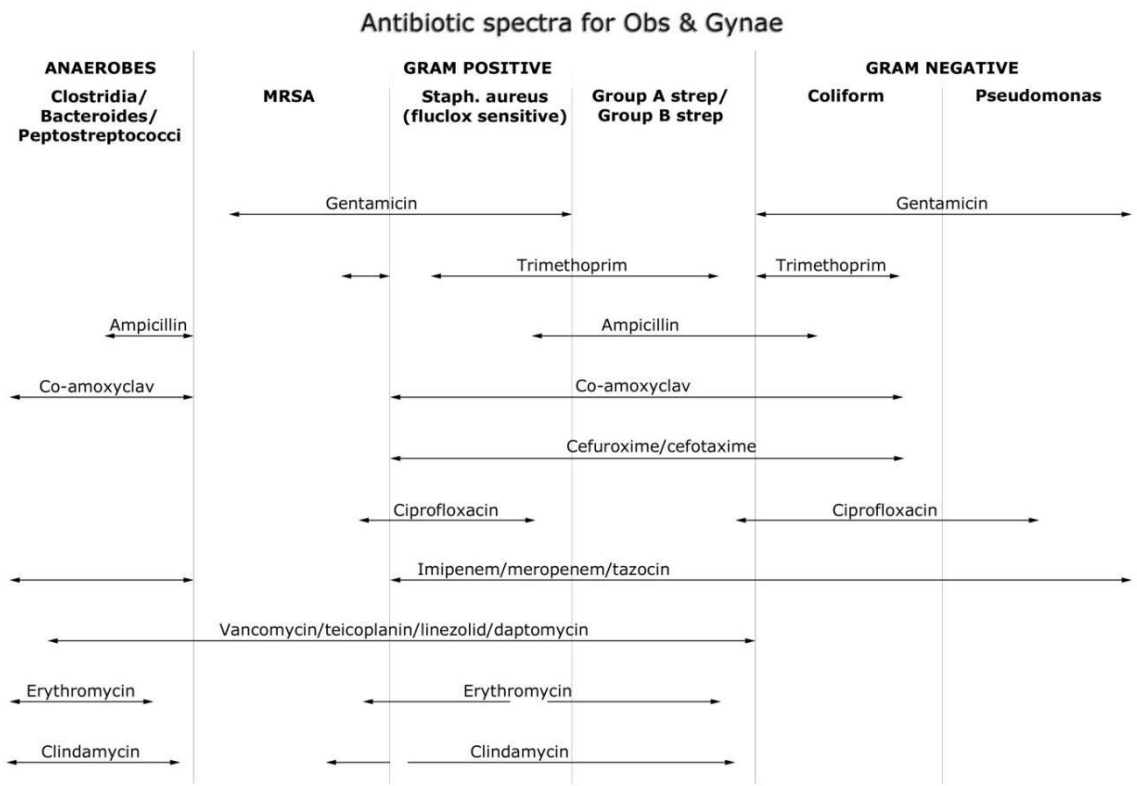
- Early recourse to care on the Labour Ward.
- All women should have a clear multidisciplinary care plan documented in the notes. This team may include the Obstetrician, Anaesthetist, Labour Ward Matron, Physician, Critical Care, Microbiologist and Neonatologist.
- All severely ill obstetric patients with sepsis should have at least a twice daily senior, multidisciplinary review as part of the consultant ward round on the labour ward and may require more frequent reviews as per individual case. Any patient who either does not improve or who is deteriorating must be discussed with a senior Obstetrician in the first instance.
- All postnatal re-admissions with infection should have a senior review within 14 hours of re-admission.
- All stable obstetric patients on intravenous antibiotics must be reviewed daily to allow a timely step down from intravenous to oral antibiotics. This should include a review of any available microbiology results and blood tests as well as clinical condition.
- Instigate cooling measures if pyrexia ≥37.5°C present – remove blankets, fan, cool compress, give intravenous paracetamol. Paracetamol is not a direct treatment for sepsis and should not take precedence over other therapeutic interventions (fluids / antibiotics).
- Maintain strict infection control precautions. Personal protective equipment and barrier nursing should be considered. Isolate patient away from other patients where possible.
- Infection Control should be informed if multi-drug resistant organisms are cultured (e.g. ESBL, MRSA etc).
- Avoid NSAIDs in sepsis due to reduce the risk of renal dysfunction. The may also impede the ability of polymorphs to fight Group A Streptococcus infection.
- Consider other imaging to assist assessment or diagnosis – chest X-Ray or Ultrasound.
- Remove the source of sepsis if possible – e.g. uterine evacuation or wound washout.
- Sepsis increases risk of thromboembolism – TED stockings should be prescribed for all and low-molecular weight heparin should be considered.

4.1 Sepsis in labour

- Women with sepsis in labour are at higher risk of severe illness and mortality.
- Continuous electronic fetal monitoring should be commenced.
- Fetal blood sampling results may be falsely reassuring and should be discussed with a senior Obstetrician.
- If the source of sepsis is thought to be the genital tract, expedite the birth.

5.0 ANTIBIOTICS

5.1 Antibiotic spectra in Obstetrics and Gynaecology



Source: RCOG Bacterial Sepsis in Pregnancy, Green-Top Guideline 64A

5.2 Antibiotics in Breastfeeding

Antibiotic	Breastfeeding
Meropenem	Present in milk but unlikely to be absorbed
Metronidazole	Significant amount in milk, avoid large single doses (> 2 grams).
Clindamycin	Amount probably too small to be harmful but bloody diarrhoea reported in one infant
Gentamicin	Considered safe
Ciprofloxacin	Amount too small to be harmful
Tazocin	Trace amounts in milk, but appropriate to use

6.0 MONITORING COMPLIANCE WITH THIS GUIDELINE

6.1 Any concern or non-compliance with this guideline that is identified through the investigation of clinical incidents, claims or complaints will be reviewed as per the Trust Policies regarding Incidents, Claims and Complaints, and may result in an audit and/or amendment to the guideline.

6.2 Relevant Policies:

- Incident reporting policy and procedure
- Claims management policy and procedure
- Policy and Procedure for the Management of Complaints, Concerns, Comments and Compliments

7.0 REFERENCES

- Antimicrobial Guidelines: Hub → A-Z → Clinical Systems → Antimicrobial Guidelines. Also available as an App.
- Cardiotocograph (CTG) continuous Electronic Fetal Monitoring (EFM) Maternity guideline
- Centre for Maternal and Child Enquiries. Saving Mothers' Lives: reviewing maternal deaths to make motherhood safer 2006-2008. CEMACE, 2011
- Knight M et al., on behalf of MBRRACE-UK. Saving Lives, Improving Mothers' Care - Lessons learned to inform maternity care from the UK and Ireland Confidential Enquiries into Maternal Deaths and Morbidity 2019-21. Oxford: National Perinatal Epidemiology Unit, University of Oxford 2023.
- Ockenden, D. Findings, conclusions and essential actions from the independent review of maternity services at the Shrewsbury and Telford Hospital NHS Trust, March 2023.
- Labour - Maternity guideline care of women in labour in all care settings
- Management of Pyrexia in Labour
- Maternity Early Warning Score (MEWS) guideline
- National Institute for Health and Care Excellence. Intrapartum care for women with existing medical conditions or obstetric complications and their babies. NICE. London NG 121, 2019.
- National Institute for Health and Care Excellence. Sepsis: Recognition, Diagnosis and Early Management. NICE. London NG 51, 2016.
- Risk Assessment & Action Plan for Suspected Maternal Sepsis
- Royal College of Obstetricians and Gynaecologists. Bacterial Sepsis In / Following Pregnancy. RCOG, London GTG 64A and 64B, 2012
- Surviving Sepsis Campaign. www.survivingsepsis.org

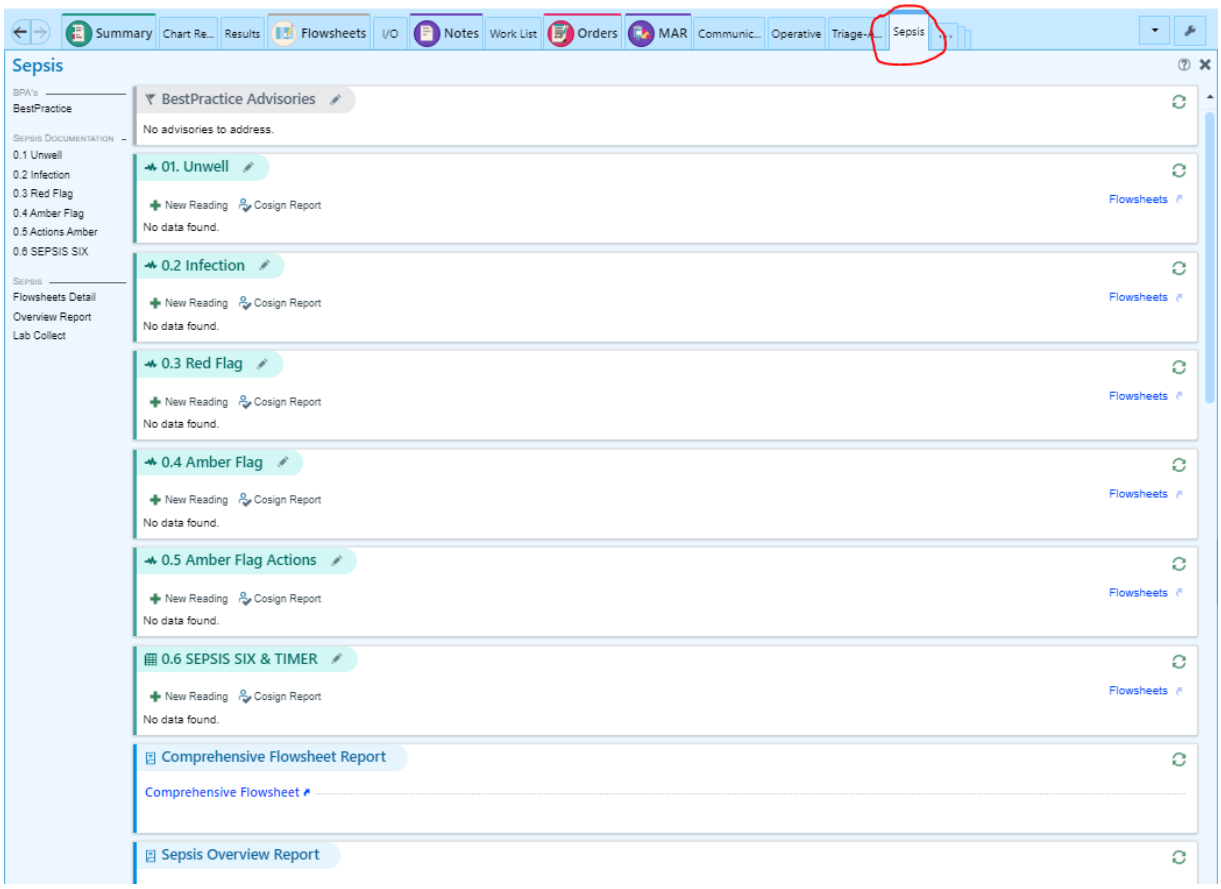
8.0 PUBLICATION DETAILS

Author of Clinical Guideline	Lead Consultant for Labour Ward
Division/ Department responsible for Clinical Guideline	Specialist Services/CWH/Maternity
Contact details	X6613
Version number	1.0 Joint guideline
Replaces version number	East v2.2 & North v4.0
Date written	East: May 2011 and North: 2014
Approving body and date approved	Maternity Guideline Group 21/02/2024
Review date	21/11/2026
Expiry date	21/02/2027
Date document becomes live	03/07/2024

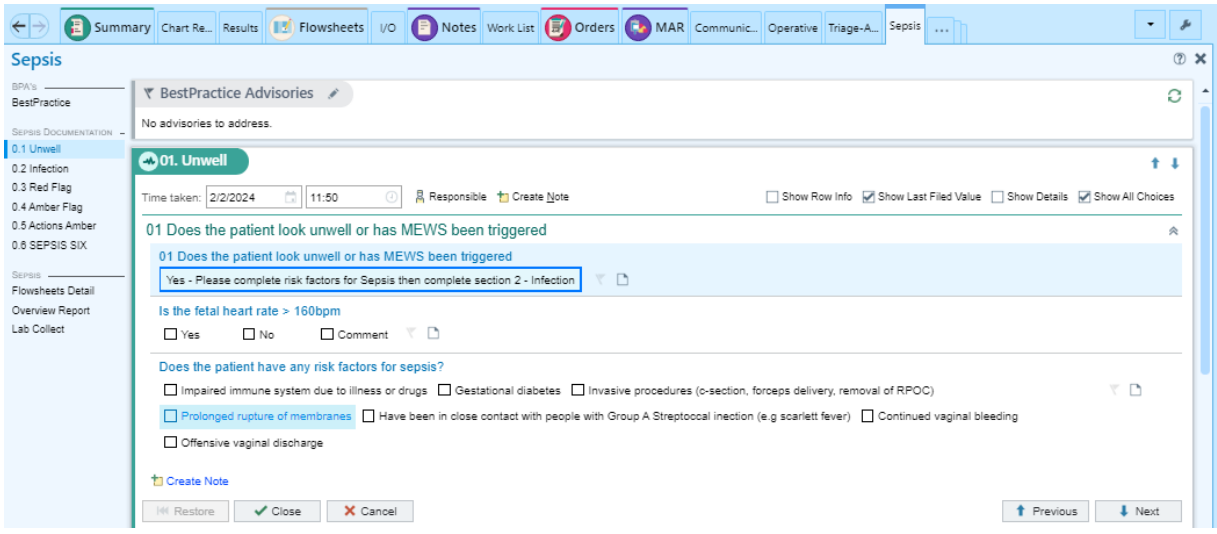
Appendix 1

Completing the OB Risk Assessment Tool

The OB Sepsis risk assessment tool is found under the Sepsis Tab in the Electronic Patient Record.



Open each section to complete



Summary | Chart Re... | Results | Flowsheets | I/O | Notes | Work List | Orders | MAR | Communic... | Operative | Triage-A... | Sepsis

Sepsis

BPA's
BestPractice

Sepsis Documentation
0.1 Unwell
0.2 Infection
0.3 Red Flag
0.4 Amber Flag
0.5 Actions Amber
0.6 SEPSIS SIX

Sepsis
Flowsheets Detail
Overview Report
Lab Collect

BestPractice Advisories
No advisories to address.

01. Unwell
New Reading | Cosign Report | Flowsheets
No data found.

0.2 Infection
Time taken: 2/2/2024 11:51 | Responsible | Create Note | Show Row Info | Show Last Filed Value | Show Details | Show All Choices

02 Could this be due to an infection?
02 Could this be due to an infection?
Yes - Please complete likely source then section 3 - Red Flag | No - Please complete Sepsis unlikely field

Likely source
☐ Chorioamnionitis ☐ Endometrosis ☐ UTI ☐ Wound Infection ☐ Influenza ☐ Pneumonia ☐ Other (state s...

If no infection - Sepsis unlikely consider other diagnoses

Create Note
Restore | Close | Cancel | Previous | Next

0.3 Red Flag
Time taken: 2/2/2024 11:53 | Responsible | Create Note | Show Row Info | Show Last Filed Value | Show Details | Show All Choices

Any RED flags present?
03 Any RED flags present? If so start SEPSIS SIX
☐ No ☐ Respiratory rate greater than 25 ☐ Sp02 < 92% without O2 ☐ Heart rate > 130 ☐ Mental status: reponds only to voice, pain or unresponsive
☐ Blood lactate >2.0* ☐ Non-blanching rash/ mottled/ ashen/ cyanotic ☐ Urine < 0.5ml/kg/hr ☐ No urine output for 18 hours

Create Note
Restore | Close | Cancel | Previous | Next

0.4 Amber Flag
Time taken: 2/2/2024 11:54 | Responsible | Create Note | Show Row Info | Show Last Filed Value | Show Details | Show All Choices

Any AMBER flags present?
04 Any AMBER flags present?
☐ No - Please complete no amber flag field ☐ Respiratory rate 21 - 24 ☐ Heart rate 100 -130 ☐ Systolic BP 91 - 100 ☐ Fetal heart > 160 ☐ Pathological CTG
☐ Prolonged SRM ☐ Recent invasive procedure ☐ Bleeding / wound infection / vaginal discharge / abdominal pain ☐ Close contact with Group A Strep
☐ Relatives concerned about mental / functional status ☐ Diabetes / immunosurpressed ☐ No urine output for 12 -18 hrs

NO AMBER FLAGS = ROUTINE CARE / CONSIDER OTHER DIAGNOSIS

Create Note
Restore | Close | Cancel | Previous | Next

0.5 Amber Flag Actions
Time taken: 2/2/2024 11:56 | Responsible | Create Note | Show Row Info | Show Last Filed Value | Show Details | Show All Choices

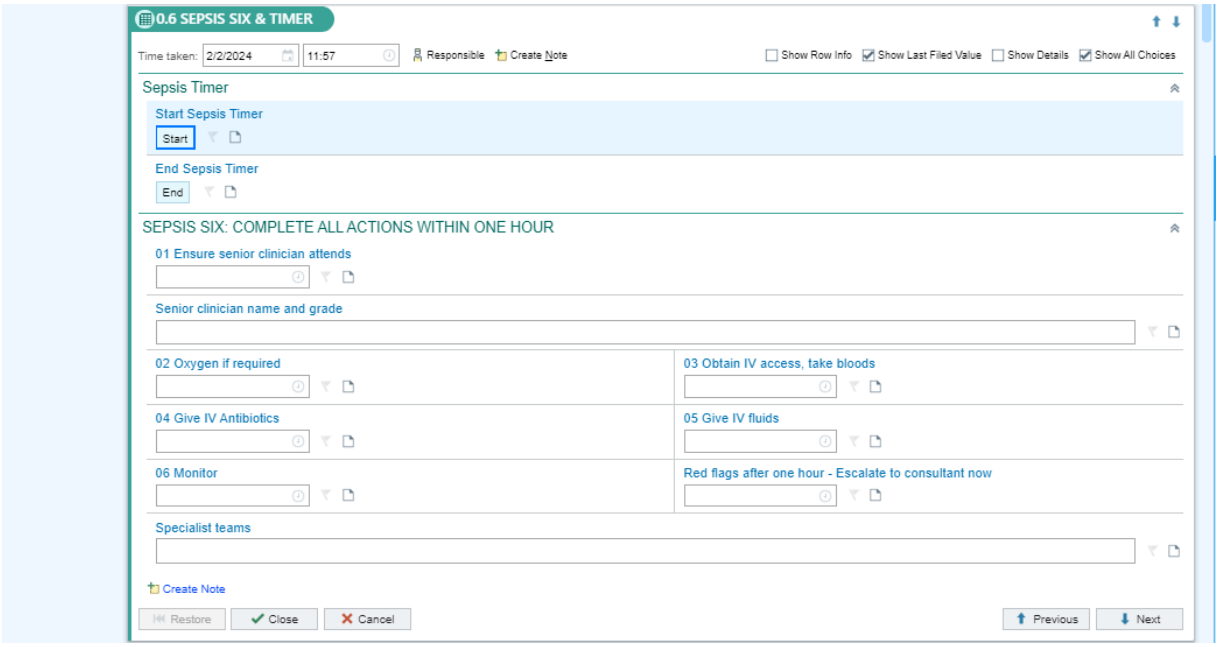
AMBER FLAGS PRESENT: Further review required
Send bloods and review results
Yes | No (reason)

Any AMBER flags then further review required: Time of review

Antibiotics required
Yes | No

Name and grade of senior review

Create Note
Restore | Close | Cancel | Previous | Next


0.6 SEPSIS SIX & TIMER
Time taken: 2/2/2024 11:57
Responsible
Create Note
Show Row Info
Show Last Filed Value
Show Details
Show All Choices
Sepsis Timer
Start Sepsis Timer
Start
End Sepsis Timer
End
SEPSIS SIX: COMPLETE ALL ACTIONS WITHIN ONE HOUR
01 Ensure senior clinician attends
Senior clinician name and grade
02 Oxygen if required
03 Obtain IV access, take bloods
04 Give IV Antibiotics
05 Give IV fluids
06 Monitor
Red flags after one hour - Escalate to consultant now
Specialist teams
Create Note
Restore
Close
Cancel
Previous
Next

Sepsis Screening, Recognition and treatment Audit Table

THINK TREAT STOP! sepsis

Date	MRN number	Any 1 Screening criteria 1. EWS>3 or if NEWS2 >5 2. Infective cause likely? 3. Is Escalation appropriate? (e.g. TEP) If yes to all 3 proceed (Yes/No)	Senior Decision maker thinks infective cause likely? Yes (proceed to sepsis 6) Or No		Sepsis 6 Bundle completed.					
					Blood Cultures (Tick box)	Urine Output monitoring (Tick box)	Fluids (Tick box)	Antibiotics within 1 hour of decision Time starts from when senior Decision maker states infective cause is likely. (please state **Yes/No** and write time in minutes)	Lactate taken (Tick box)	O2 Given (Tick box)
			YES	**NO**				Yes /NO Mins		
			YES	**NO**				Yes /NO Mins		
			YES	**NO**				Yes /NO Mins		
			YES	**NO**				Yes /NO Mins		
			YES	**NO**				Yes /NO Mins		
			YES	**NO**				Yes /NO Mins		

Date	MRN number	Any 1 Screening criteria 1. EWS>3 or if NEWS2 >5 2. Infective cause likely? 3. Is Escalation appropriate? (e.g. TEP) If yes to all 3 proceed (Yes/No)	Senior Decision maker thinks infective cause likely? Yes (proceed to sepsis 6) Or No		Sepsis 6 Bundle completed.					
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			YES	**NO**				Yes /NO Mins		
			YES	**NO**				Yes /NO Mins		
			YES	**NO**				Yes /NO Mins		
			YES	**NO**				Yes /NO Mins		
			YES	**NO**				Yes /NO Mins		
			YES	**NO**				Yes /NO Mins		
			YES	**NO**				Yes /NO Mins		
			YES	**NO**				Yes /NO Mins		